4.4 Congruence and Transformations

Speak **L**ike **A** **M**athematician:

congruence transformation:

composition of transformations:

rigid motion:

Congruence Transformations

A transformation that preserves length and angle measure is a _____ _____.

Two figures are _____ if and only if there is a congruence transformation that maps one onto the other.

The three rigid motions are _____, _____, and _____.

Composition Theorem

The composition of two or more rigid motions is a _____ _____.

So a figure and its final image after ANY number of rigid motions are still _____.

Reflections in Parallel Lines

If lines m and n are _____, a reflection in m followed by a reflection in n is a _____.

The translation is _____ to both lines, and its length is _____ the distance between them.

Reflections in Intersecting Lines

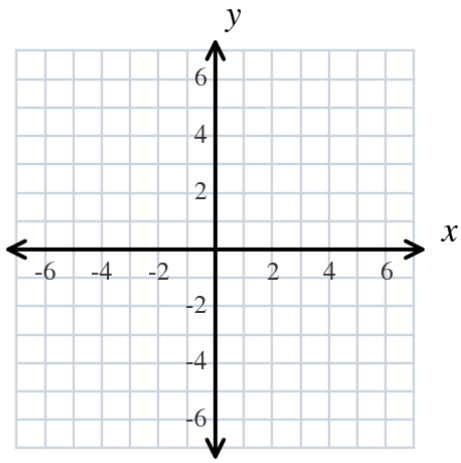
If lines m and n intersect at point P, a reflection in m followed by a reflection in n is a _____ about _____.

The angle of rotation is _____ the angle between m and n.

Example 1 – Name the Congruence Transformation

$\triangle ABC$ has A(1, 2), B(4, 3), and C(3, −1). Its image $\triangle A'B'C'$ has A′(1, −2), B′(4, −3), and C′(3, 1).

4.4 Congruence and Transformations



a) Graph both triangles.

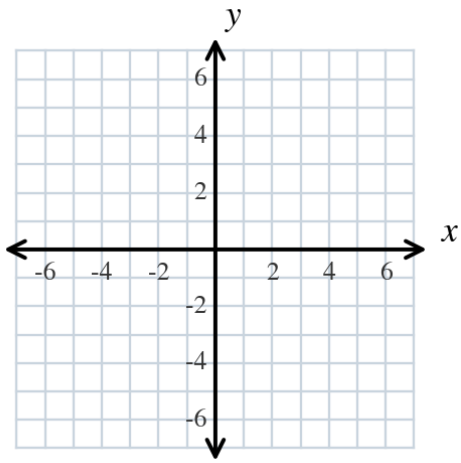
b) Name the congruence transformation.

c) Write its coordinate rule.

d) Why is $\triangle ABC \cong \triangle A'B'C'$?

Example 2 – Describe a Composition

$P\bar{Q}$ has P(2, 3) and Q(5, 1). After a reflection in the x-axis followed by the translation $(x, y) \rightarrow (x - 6, y)$, find the final image.



a) Find the coordinates after the reflection.

b) Find the coordinates after the translation.

c) Is the composition a congruence transformation? Explain.

Example 3 – Reflections in Two Parallel Lines

Lines m and n are vertical, with m: x = 1 and n: x = 4.
Reflect a figure in m, then reflect the result in n.

a) Follow the point A(0, 0). Where is it after each reflection?

b) Write ONE rule for the composition.

c) How does the 6 relate to the distance between m and n?

Example 4 – Reflections in Two Intersecting Lines

Reflect a figure in the line y = x, then reflect the result in the y-axis.

a) Follow B(2, 3) through both reflections.

b) What single transformation does this match?

c) The two lines meet at 45°. How does that explain the 90°?